WORK.GROUP

Smart Curriculum Activity and Attendance Management System

Design Thinking for Innovation & Entrepreneurship (DTIE) | End Semester 2026–27

Section: 16 | Project Batch: A | Faculty Guide: Sudhakar sir (official full name pending)

Team Members

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1. Executive Summary

Work.Group is a browser-based educational management prototype intended to consolidate class administration, attendance, academic activities and communication. The supplied HTML includes teacher, student, co-star and administrator interfaces. Its menus include classes, attendance, exam data, homework, notices, messaging, meetings and timetables. This document describes the prototype and a proposed path toward deployment; it does not claim that the application has been tested with real institutions.

2. Vision and Mission

Vision: Provide one accessible digital workspace for routine academic coordination.

Mission: Reduce fragmented manual records, make attendance and academic information easier to access, and improve communication between teachers, students and administrators.

3. Problem Statement

When attendance, timetables, homework and performance records are managed in separate notebooks, spreadsheets or messaging channels, teachers must duplicate work and students may miss updates. A single interface could organize these workflows and make class information easier to retrieve.

4. Research / Facts / Market & Industry Study

Background: Educational management products commonly address attendance, academic records and communications. Work.Group explores combining these activities in one interface. The supplied project

files contain no survey results, quantified market sizing or verified customer interviews. Such evidence must be collected before claiming measurable demand or efficiency improvements.

Research status: No formal teacher/student interviews, surveys, or independent market research have been conducted. Proposed next step (not completed): interview teachers and survey students to evaluate need and usability.

5. Target User / Customer

- Teachers: manage classes, students, attendance, exams, notices and meetings.
- Students: view attendance, exam information, timetable, messages and homework.
- Institution administrators: review classes, teachers, students, homework, timetables and attendance history.

6. Competitors / Benchmarks

Comparison categories include school ERP platforms, learning management systems and spreadsheet-plus-messaging workflows. Evaluate alternatives on ease of use, cost, attendance entry, student access, communication, reporting and privacy. No named competitor pricing or performance claims have been verified in the supplied files.

7. Risks and Gaps

- Data protection: student records require access controls, secure storage and an appropriate privacy policy.
- Prototype persistence: the HTML contains browser localStorage usage; local browser data is not a substitute for centralized, backed-up institutional records.
- Multi-user reliability: synchronization, authentication and server-side authorization require verification before deployment.
- Adoption: teachers and students need training, accessible layouts and reliable connectivity.
- Validation: user testing, security testing and measured outcomes have not been supplied.

8. Concept / Idea / Prototype / Solution

Development status: working prototype, partially tested. The supplied HTML demonstrates role-oriented screens and client-side functionality. Team-confirmed working modules: attendance, curriculum/activity tracking, timetable, messaging and homework. Full deployment and comprehensive tests are not claimed.

- Class administration: create and review classes; add, search and manage student records.
- Attendance: daily attendance entry, absent-student views, attendance history and charts.
- Academic activities: homework, class notices, exam data and student performance views.
- Planning: class timetables, weekly timetable editing, holidays and academic year sections.
- Communication: teacher/student chat, messages and meeting scheduling.
- Presentation: responsive card-based design and dark-mode styling.

9. Architecture / User Flow

Proposed logical flow: User → role selection/login → role dashboard → feature module → record retrieval/update → dashboard or report.

Current prototype: HTML, CSS and JavaScript with browser-based data storage. This does not establish secure, synchronized multi-device operation; a hosted database is a possible future improvement.

Suggested storyboard: (1) Teacher logs in and opens a class. (2) Teacher records attendance. (3) Student opens attendance history. (4) Administrator reviews summary charts. Screenshots of the running application can be inserted here for final submission.

10. Lean Canvas Model

Problem: Fragmented attendance and academic activity tracking.

Customer Segments: Teachers, students, school/college administrators.

Unique Value Proposition: One interface for classroom records and communication.

Solution: Role-specific dashboards, attendance, homework, exams, timetable and messaging.

Channels: Institution pilots, faculty demonstrations and student onboarding.

Revenue Streams: Possible institution subscriptions or licensing; not validated.

Cost Structure: Hosting, development, security, maintenance and training.

Key Metrics: Active users, attendance entry time, update visibility, error rate.

Unfair Advantage: Not yet demonstrated; must be established through usability and adoption.

11. Capital (Financial)

Illustrative planning budget only, not actual expenditure: domain ₹1,000–₹2,000/year; hosting ₹3,000–₹12,000/year for a small pilot; miscellaneous testing and deployment ₹2,000–₹5,000. Indicative first-year out-of-pocket total: ₹6,000–₹19,000, excluding development labor and any paid external services. Actual quotes and architecture may change these amounts.

12. Roadmap / Timeline

- Weeks 1–2: Interview teachers/students; confirm requirements and permissions.
- Weeks 3–4: Refine role dashboards and user journeys; fix usability issues.
- Weeks 5–7: Add secure backend, database and account authorization.
- Weeks 8–9: Test attendance, reports, timetables and messaging with sample data.
- Weeks 10–11: Pilot with one class; collect error and usability feedback.
- Week 12: Improve documentation, security controls and final demonstration.

13. Testing and Evaluation Plan

- Functional: teacher records attendance and student sees the correct entry.
- Functional: teacher posts homework and student views the correct class notice.
- Security: student cannot access another student’s private records.
- Reliability: verify data remains correct across browsers and sessions once backend is added.
- Usability: measure task completion and collect feedback from representative users.

14. Conclusion

Work.Group demonstrates an integrated concept for academic activity and attendance management. Its interface covers several teacher, student and administrative workflows. The next priority is to validate the problem with users and move from a browser prototype toward a secure, tested multi-user system.

15. Source Materials and Submission Notes

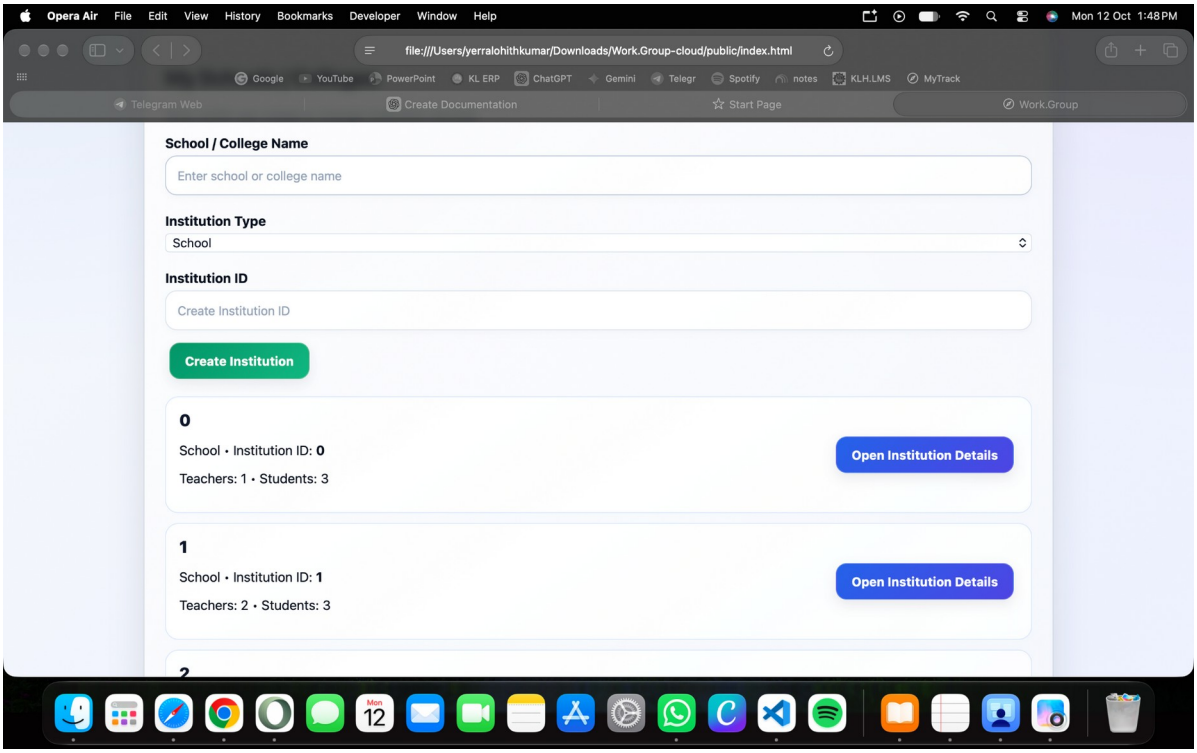
Primary source 1: DTIE Pitch Deck Template for End Sem-2026-27.pptx (required outline).
Primary source 2: index(20261012-081338).html (Work.Group prototype).

Before submission: fill team and faculty details; add original screenshots; collect research citations; replace estimated capital with actual costs; verify all features in a live demonstration.

Prototype Screenshots and Evidence

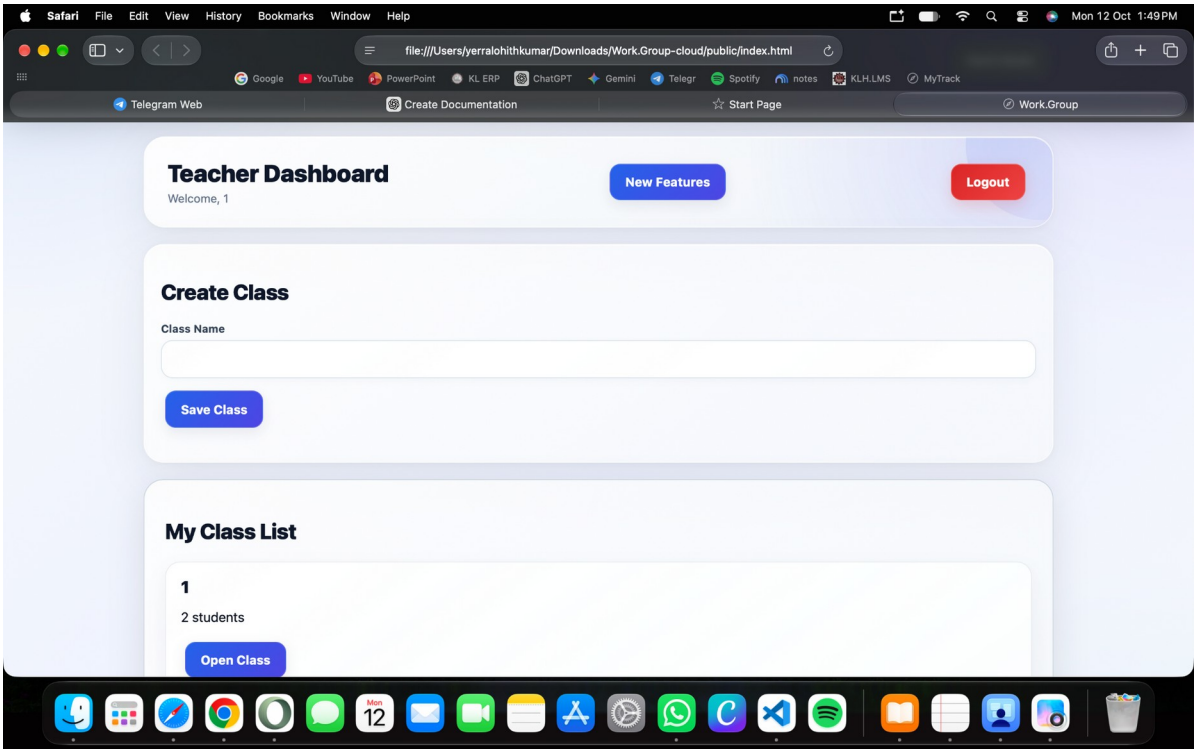
The following screenshots were supplied by the project team on 12 October 2026. They demonstrate visible interface screens, not comprehensive verification of all application functions.

Figure 1. Institution administration: institution creation form and institution records.



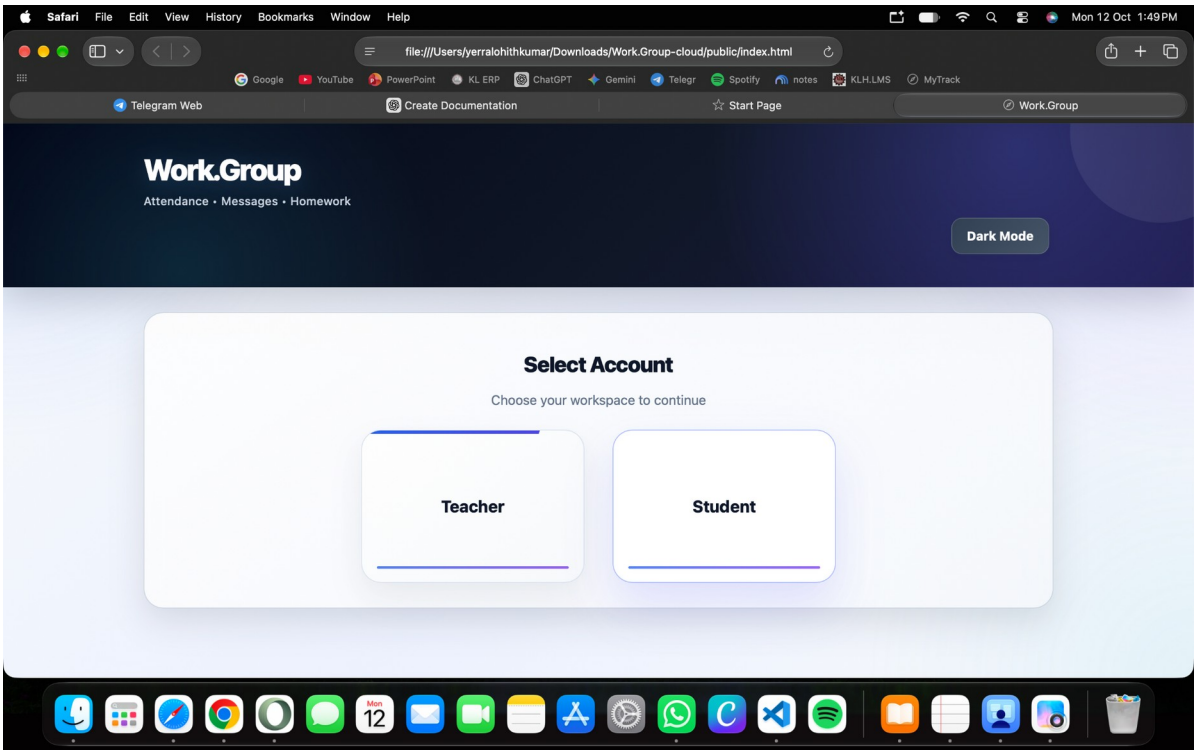
Source: team-supplied screenshot of local Work.Group prototype.

Figure 2. Teacher dashboard: create-class form and class listing.



Source: team-supplied screenshot of local Work.Group prototype.

Figure 3. Home page: teacher/student role selection and theme control.



Source: team-supplied screenshot of local Work.Group prototype.

Submission Notes and Evidence Gaps

- Faculty guide: “Sudhakar sir” is a provisional label; verify full official name and designation before final submission.
- No formal interviews, user surveys or independently sourced market statistics have been provided.
- Testing is partial; no formal test cases, pass rates, or deployment results have been supplied.
- Actual development expenses: ₹0 using free tools; future operational budget remains uncalculated.
- Screenshots demonstrate home, teacher and institution screens; add screenshots of attendance, curriculum, timetable, messages and homework if available.
- The prototype uses browser storage; cloud synchronization and production-grade access controls should not be represented as implemented.